

Project Progress Report: Fusion-Baukasten

Date: July 17, 2026

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Overview

This report provides a high-level summary of the foundational work completed for the **Fusion-Baukasten** project.

Before we can build the visual screens that users click on, we first had to construct the invisible "plumbing" of the software. Without this foundation, the application cannot securely store data, process user logins, or connect to the AI. Here is exactly what was built over the last phase and why it is critical for the project's success.

What We Built & Why It Matters

1. The Code Vault (Repository & Version Control)

What was done: We set up a professional GitHub repository. We also cleaned up the file structure to separate heavy design files from the raw application code.

Why we needed it: Think of this as a highly secure, digital vault for our blueprints. It tracks every single line of code written. If we ever make a mistake, we can instantly "rewind" time to a working version. It also ensures that if more developers join the team later, they have a clean, organized workspace to contribute to.

2. The Digital Filing Cabinet (Database Architecture)

What was done: We designed and deployed a complete PostgreSQL database. Specifically, we built 14 distinct "tables" (data structures) tailored to our exact business logic.

Why we needed it: When a user types a project goal or the AI generates a response, that data needs to be saved permanently. We built specialized "filing drawers" for:

- **User Accounts:** To safely store who is registered.
- **Projects & Tasks:** To keep track of which team is working on what.
- **AI Conversations:** To remember the chat history so the AI doesn't lose context.

3. The Brain & The Bouncer (Backend Server & Security)

What was done: We built the core "Backend" server using Python (FastAPI). Inside this server, we implemented an industry-standard security and authentication system.

Why we needed it: If the database is the filing cabinet, the Backend server is the librarian—it's the only thing allowed to read or write files. We added a "Bouncer" (JWT Authentication and password encryption) to ensure that hackers cannot steal user data, and that users can only see their own private projects.

4. The Storefront Foundation (Next.js Frontend)

What was done: We initialized the "Frontend" application (Next.js) and configured the design system (TailwindCSS) to perfectly match the brand colors and typography from our Figma designs.

Why we needed it: The frontend is the actual website the user sees. By setting this foundation up correctly

now, we guarantee that when we start building the buttons and forms next week, they will automatically look premium, load instantly, and work flawlessly on all devices.

Next Steps

Now that the invisible foundation is rock-solid and secure, we are moving into the highly visual phase.

Our immediate next step is **Feature 1: The Login Screen & Dashboard**. We will build the visual screens where users type in their passwords, and connect those screens directly to the "Bouncer" and "Filing Cabinet" we just finished building.